Overall

Overall — 1.1 — vulcan-vulkan-mesa-git

Score	Time	Cycles	GPU	GPU Frequency	Process	OS Release
1.1	112.12s, 1000.00 s	112.12 Gcy, 1000.00 Gcy	Kimmie Gallium (RTX A4000)	1.00 GHz	Linux-6.8.0-106-generic	Linux 6.8.0-106-generic

Summary | **Details** | Graphics | Certificate | Compatibility | Raw | Remarks

Hardware | Software | Configuration | Performance | Power | Thermal | Limits | Graphics | Errors

Overall

Overall Score — 1.1

Overall score of the Threadripper Pro workstation, high-end workstation, or other demanding workstations. Typical workstations will score lower, and systems with less than 16 GB RAM will score significantly lower.

Overall Performance

Overall performance of the Threadripper Pro workstation, high-end workstation, or other demanding workstations. Typical workstations will score lower, and systems with less than 16 GB RAM will score significantly lower.

Overall Stability

Overall stability of the Threadripper Pro workstation, high-end workstation, or other demanding workstations. Typical workstations will score lower, and systems with less than 16 GB RAM will score significantly lower.

Overall Compatibility

Overall compatibility of the Threadripper Pro workstation, high-end workstation, or other demanding workstations. Typical workstations will score lower, and systems with less than 16 GB RAM will score significantly lower.

Overall Security

Overall security of the Threadripper Pro workstation, high-end workstation, or other demanding workstations. Typical workstations will score lower, and systems with less than 16 GB RAM will score significantly lower.

Overall Remarks

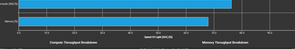
Overall remarks of the Threadripper Pro workstation, high-end workstation, or other demanding workstations. Typical workstations will score lower, and systems with less than 16 GB RAM will score significantly lower.

GPU Throughput

GPU Throughput — Computer and Memory are well balanced. To reduce runtime, both compute and memory traffic must be reduced. Use the Roofline chart and the Memory Throughput chart to identify the bottleneck.

Vulkan Version — This score is based on Vulkan API tests. Vulkan API performance is the device's raw 3 performance. Vulkan API performance does not necessarily correlate to application performance. Vulkan API performance may differ from OpenGL performance.

GPU Throughput



Geekbench GPU (OPENCL)

Compute Throughput Breakdown

64 - Gauss Jordan Elim	76.75	1.1 - Matrix Transpose Shared(Int4) (T)
63 - Matrix Inverse(Int4) (T)	75.75	2.1 - Matrix Multiply(Int4) (T)
65 - Matrix Trace Trace Vector Int4 (T)	67.75	1.1 - Vector Reduce(Int) (T)
62 - Matrix Determinant(Int4) (T)	67.75	1.1 - Vector Reduce(Int4) (T)
66 - Matrix Trace Vector Int4 (T)	67.75	1.1 - Vector Reduce(Int8) (T)
61 - Matrix Cofactor(Int4) (T)	67.75	1.1 - Vector Reduce(Int16) (T)
60 - Mat Add Int4 (T)	67.75	1.1 - Vector Reduce(Int64) (T)
59 - Mat Mul Int4 (T)	67.75	1.1 - Vector Reduce(Int8 Accumulate) (T)
58 - Mat Vec Mul Int4 Vector Int4 (T)	67.75	1.1 - Vector Reduce(Int16 Accumulate) (T)
57 - Mat Vec Mul Int4 Scalar Int4 (T)	67.75	1.1 - Vector Reduce(Int64 Accumulate) (T)
56 - Mat Vec Matrix Scalar Int4 (T)	67.75	1.1 - Vector Reduce Fast(Int) (T)
55 - Mat Vec Matrix Vector Int4 (T)	67.75	1.1 - Vector Reduce Fast(Int4) (T)
54 - Matrix Transpose(Int4) (T)	67.75	1.1 - Vector Reduce Fast(Int8) (T)
53 - Transpose Vector Int4 (T)	67.75	1.1 - Vector Reduce Fast(Int16) (T)
52 - Transpose Scalar Int4 (T)	67.75	1.1 - Vector Reduce Fast(Int64) (T)
51 - Matrix Add Int4 (T)	67.75	1.1 - Sequence Fill(Int4) (T)
50 - Int4 Scalar Mad (T)	67.75	1.1 - Sequence Fill(Int8) (T)
49 - Int4 Vector Mad (T)	67.75	1.1 - Vector Bit Flip Test Vector Int4 (T)
48 - Int4 Vector Mad Sat (T)	67.75	1.1 - Vector Bit Flip Test Vector Int8 (T)
47 - Int4 Vector Mad Hi (T)	67.75	1.1 - Vector Bit Flip Test Vector Int16 (T)
46 - Int4 Vector Mad Hi Sat (T)	67.75	1.1 - Vector Bit Flip Test Vector Int64 (T)
45 - Flop Vector Scalar Int4 (T)	67.75	1.1 - Vector Bit Flip Test Vector Int4 (T)
44 - Flop Vector Vector Int4 (T)	67.75	1.1 - Sequence Fill(Int64) (T)

Memory Throughput Breakdown

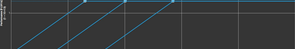
1.1 - Vector Fill(Int4) (T)	67.75
1.1 - Vector Fill(Int8) (T)	67.75
1.1 - Vector Fill(Int16) (T)	67.75
1.1 - Vector Fill(Int64) (T)	67.75
1.1 - Vector Fill Fast(Int4) (T)	67.75
1.1 - Vector Fill Fast(Int8) (T)	67.75
1.1 - Vector Fill Fast(Int16) (T)	67.75
1.1 - Vector Fill Fast(Int64) (T)	67.75
1.1 - Sequence Fill(Int4) (T)	67.75
1.1 - Sequence Fill(Int8) (T)	67.75
1.1 - Sequence Fill(Int16) (T)	67.75
1.1 - Sequence Fill(Int64) (T)	67.75

Floating Point Operations Roofline

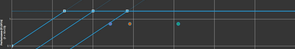
Floating Point Operations Roofline (Double Precision)

Floating Point Operations Roofline (Half Precision)

Floating Point Operations Roofline (Single Precision)



Tensor Core Operations Roofline



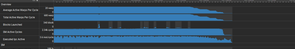
Arithmetic Intensity (FP16/Int4)

Kernel	Arithmetic Intensity (FP16)	Arithmetic Intensity (Int4)	Tensor Core Operations (Count)	Other Operations (Count)
4x4 fp16 fma fp16 Operands vf	0	0	0	65,536 - Flop Vector Scalar Int4 (T)
4x4 fp16 fma fp16 Operands vf	0	0	0	65,536 - Flop Vector Vector Int4 (T)
4x4 fp16 fma fp16 Operands vf	0	0	0	65,536 - Int4 Scalar Mad (T)
4x4 fp16 fma fp16 Operands vf	0	0	0	65,536 - Int4 Vector Mad (T)
4x4 fp16 fma fp16 Operands vf	0	0	0	65,536 - Int4 Vector Mad Sat (T)
4x4 fp16 fma fp16 Operands vf	0	0	0	65,536 - Int4 Vector Mad Hi (T)
4x4 fp16 fma fp16 Operands vf	0	0	0	65,536 - Int4 Vector Mad Hi Sat (T)
16x16 fp16 fma fp16 Operands vf	0	0	0	1,179,648 - FP16 Flop Vector (T)
16x16 fp16 fma fp16 Operands vf	0	0	0	1,179,648 - FP16 Flop Vector Fma (T)
16x16 fp16 fma fp16 Operands vf	0	0	0	1,179,648 - FP16 Flop Vector Sat (T)
16x16 fp16 fma fp16 Operands vf	0	0	0	1,179,648 - FP16 Flop Vector Sat Fma (T)
16x16 fp16 fma fp16 Operands vf	0	0	0	1,179,648 - FP16 Flop Vector Hi (T)
16x16 fp16 fma fp16 Operands vf	0	0	0	1,179,648 - FP16 Flop Vector Hi Fma (T)
16x16 fp16 fma fp16 Operands vf	0	0	0	1,179,648 - FP16 Flop Vector Hi Sat (T)
16x16 fp16 fma fp16 Operands vf	0	0	0	1,179,648 - FP16 Flop Vector Hi Sat Fma (T)
16x16 fp16 fma fp16 Operands vf	0	0	0	1,179,648 - Int4 Vector Mad (T)
16x16 fp16 fma fp16 Operands vf	0	0	0	1,179,648 - Int4 Vector Mad Sat (T)
16x16 fp16 fma fp16 Operands vf	0	0	0	1,179,648 - Int4 Vector Mad Hi (T)
16x16 fp16 fma fp16 Operands vf	0	0	0	1,179,648 - Int4 Vector Mad Hi Sat (T)

File Systems

File Systems — Performance for random write/read/delete over files stored on disk. Raid is controller and multi-threaded dependent. Raid is controller and multi-threaded dependent. Changes over time can indicate a failing drive.

Maximum Sustain Write (Mb/sec)



Compute Shaders

Compute Shaders — Compute shaders are a simpler way to access the power of the underlying hardware(s). GMS, including the software rasterizer, uses GPU cycles for utilization of each available shader. Platforms with very high utilization might exhibit lower overall performance.

Compute Shaders Analysis — Compute shaders are a simpler way to access the power of the underlying hardware(s). GMS, including the software rasterizer, uses GPU cycles for utilization of each available shader. Platforms with very high utilization might exhibit lower overall performance.

Compute Shaders Analysis

Kernel	Dispatches	Average Time	Minimum Time	Maximum Time	Total Time
4x4 fp16 fma fp16 Operands vf	1	0.00 ms	0.00 ms	0.00 ms	0.00 ms
16x16 fp16 fma fp16 Operands vf	1	0.00 ms	0.00 ms	0.00 ms	0.00 ms